

Engineering Design Specifications for a Rocket with Payload Deployment

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October 19th, 2025

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This design project aims to pioneer how payload deployments are executed in rocketry by creating a sequential payload deployment (SPD) system that operates during and/or after flight for high-powered rockets. Though the scope of the project will be relatively small given the experimental equipment the team has access to, the deployment system that will be developed will be applicable to launch vehicles of any size. For anyone interested in deploying multiple payloads in a single flight either before or after landing, this system is applicable. This level of utility results in potential primary end users in the space exploration and national defense sectors. NASA – as well as any contractors working alongside or under NASA in space exploration programs – in their ultimate goal to explore deeper regions of space would be able to deliver more supplies over various distances for astronauts stationed on the Moon, Mars, or any other celestial bodies. Lockheed Martin, Northrop Grumman, General Dynamics, and any other defense company that has divisions involved with aerospace development could develop vehicles that deliver payloads to infantry, or any personnel present on other celestial bodies, over a vast area with this technology. Even within hobby rocketry, anyone intending on testing multiple payloads can do so at once with one launch vehicle instead of multiplying the project's cost by making several rockets. This SPD system minimizes cost by ensuring efficacy in various mission objectives.

User Wants and Needs

As engineers the goal is to design, manufacture, and implement a product that provides a practical solution to real world problems. Understanding what issues customers and stakeholders would like to resolve is directly linked to the success of the rocket system because their needs

define the purpose and performance expectations of the design. To support the design and development process, customer wants/needs were collected, organized, and prioritized to guide technical specification development. These requirements reflect both mission performance objectives and operational considerations, ensuring the rocket and its subsystems meet end-user needs for safety, functionality, reliability, and usability.

The top priority is ensuring a safe and stable flight trajectory and payload deployment throughout the entire flight. Customers expect the vehicle to launch, ascend, and descend without uncontrolled motion or instability. This requirement drives design considerations such as structural integrity, aerodynamic stability, and flight software reliability from the electric components like sensors. A safe flight protects not only the payloads but also customers present at the launch site, equipment, and the test environment.

Having the ability to carry and safely deploy multiple payloads at different altitudes is the primary mission function for the high-powered rocket. Users require that the system can hold multiple payloads while maintaining stable flight characteristics. Payload deployment must be reliable, timed correctly, and avoid interference with other subsystems. This affects the payload bay design, deployment mechanism, telemetry data, and power management strategies. Additionally, ensuring a recoverable rocket is essential for post-flight data collection.

Affordability is a key constraint for many educational and research rocketry projects. The design must balance performance goals with budgetary limitations, ensuring that components are cost-effective, accessible, and maintainable. This influences choices in materials, manufacturing processes, avionic systems, and telemetry solutions. Affordability does not override safety or functionality but acts as a critical design driver.

Customers expect the rocket to maintain good aerodynamic efficiency, which reduces drag and enhances altitude performance. This requires careful shaping of the body, fin alignment, and mass distribution. At the same time, the rocket must be easy to transport, implying lightweight construction, modularity, and compact storage. This dual requirement encourages designs that are structurally efficient and practical for field operations.

Real-time and post-flight data are essential for both performance evaluation and vehicle tracking in flight. Users require accurate in-flight data acquisition to monitor critical variables such as acceleration, velocity, altitude, orientation, and position. In parallel, onboard data storage ensures that information is preserved even if the telemetry link is interrupted so one can still review the information after the rocket is recovered. Location and altitude tracking accuracy are important for recovery and for the payload release mechanism to release payloads at set altitudes. Real-time tracking allows the ground team to monitor the rocket's position and other flight data during and post flight. These needs drive the selection of telemetry systems, antennas, and ground station configurations.

Users also value operational simplicity for rocket assembly and payload loading before launch. The rocket should be modular to allow for easy repairs and reduction in assembly time. Each component should be reasonably accessible without the need of heavy machinery in case small modifications need to be made pre-flight.

The requirements identified provide a structured foundation for system design by prioritizing safety, mission functionality, and recoverability. The design team ensures the rocket system not only meets technical performance goals but also delivers an operationally effective, affordable, and user-friendly solution. By clearly understanding and addressing the customer or

stakeholders core issues the engineering team ensures that resources are allocated to what matters most so that mission objectives can be achieved.

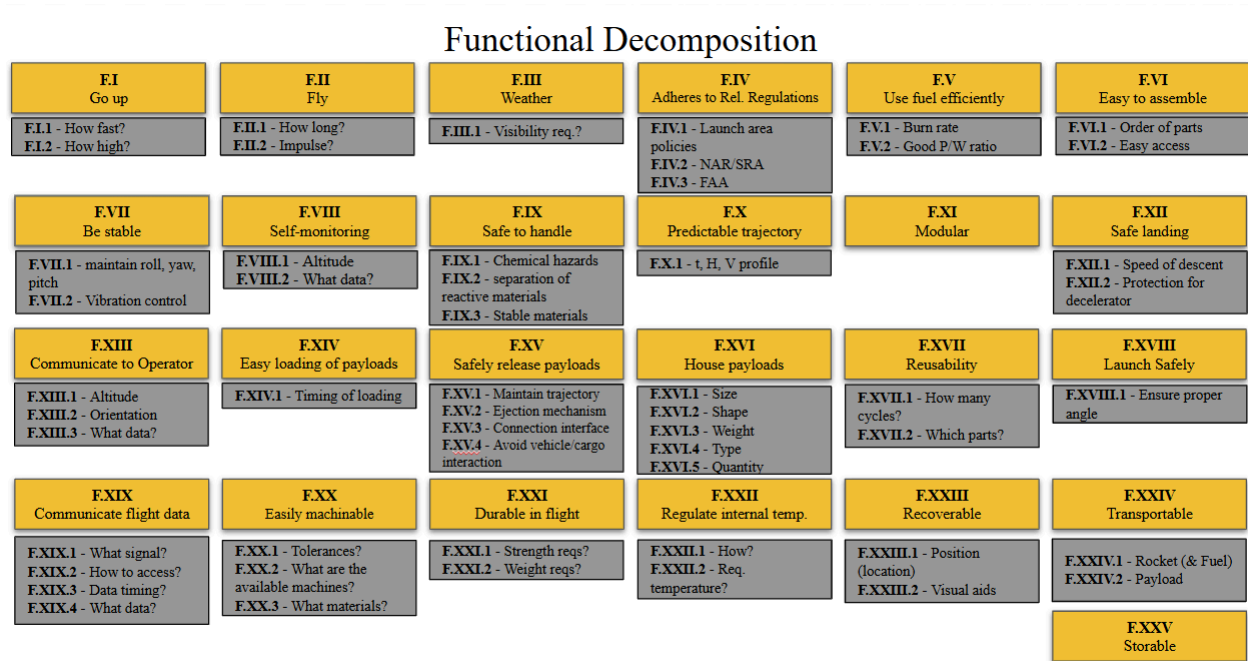


Figure 1. Functional Decomposition

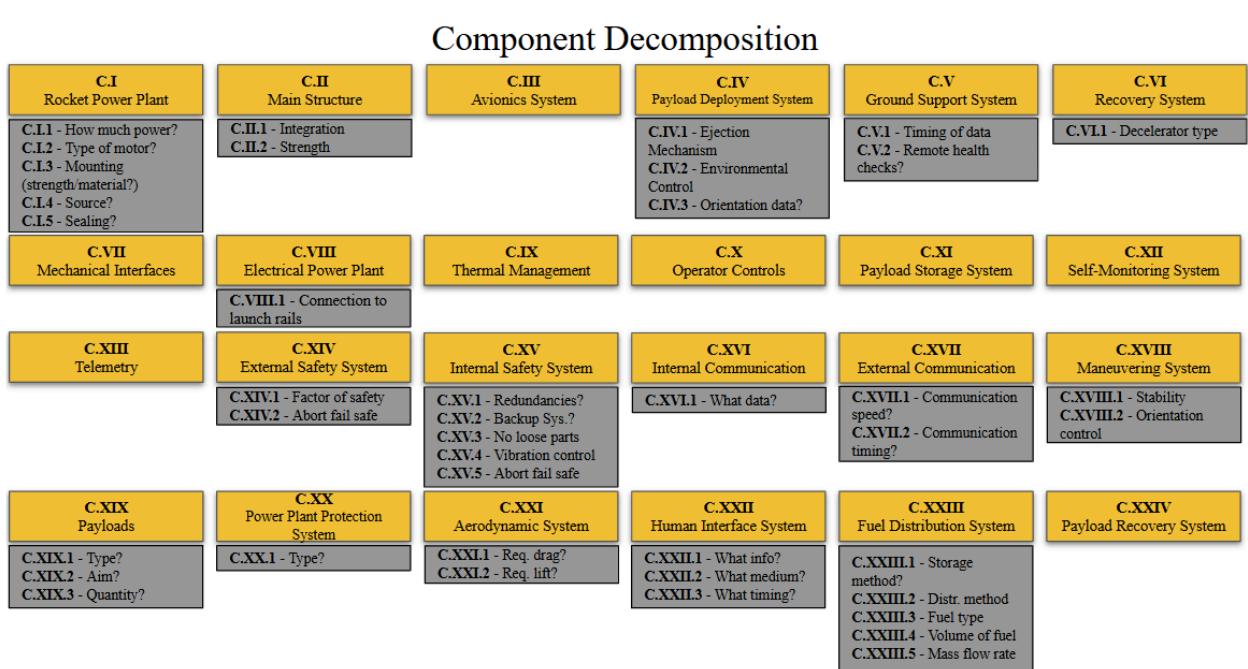


Figure 2. Component Decomposition

Functional and Component Decompositions were constructed (Figures 1&2) to determine the necessary functions and generalized components of the design. They serve as the preliminary tool to connect customer wants and needs to engineering specifications.

Competitor Benchmarks

Before finalizing the design of the group's rocket, it is essential to conduct a thorough competitor analysis to understand existing high-power rocket kits and the engineering trade-offs they represent. This section examines several pre-made rocket kits that fall mostly within the expected range of the group's rocket's conceptual design. By comparing and noting these rocket kits across key metrics such as cost, materials, manufacturing processes, total weight, and deployment mechanisms, we can identify performance benchmarks that will guide our design decisions. Each competitor breakdown will include relevant technical specifications, a list of system components, and comments regarding the payload bay configurations and recovery systems. The insights gained from this process will be used to develop a clear list of engineering design requirements for the group's final prototype.

Competitor: EZI-65 (4.00" diameter)

The first competitor found was the high-power rocket kit from Apogee Rockets called ESI-65 [1]. This rocket kit is \$145.10 and consists of the premium body tubes, plastic nose cone, plywood bulkhead, pre-cut fin slots, pre-cut plywood fins, a motor mount, and a simple recovery system.

Body Tubes

The body tubes in this kit are thick-walled cardboard meant for extra strength. The kit comes with the main body tubes, coupler, motor mount, and payload bay. The body tube also has

pre-cut fin slots. The body tube has “through-the-wall fin mounting” where the ends of the fins slip through the slots in the body tube and mount directly to the engine mount tube. The body tubes can withstand the thrust of the motor and are compatible with an L1 motor. The rocket kit also has a 54mm motor mount, so it can be used for an L2 certification. The total length of the rocket is 57.75” with a max diameter of 4.00”, and it weighs around 2.56lbs. The payload bay internal diameter is 3.8” and the payload bay internal length is 8.75”.

Nose Cone

The kit comes with a 12.5” long plastic nose cone that has an outer attachment hook for the shock cord to connect to. They note in the kit that the center of pressure (CP) location is 53.87” from the tip of the nose cone.

Plywood Parts

There are a few plywood parts in the kit, including: three fins, a plywood bulkhead, and three centering rings. The three fins are made from 1/8- inch thick, high-density, premium plywood and are also pre-cut to insert into the fin slots on the body tube. The span of the fins is 11.50”. The centering rings are used to stabilize the motor mount in the body tube. The rear centering ring epoxies to the fin tabs, which increases the strength of the fins.

Recovery System

The kit also contains most of the necessary recovery components. It includes a long nylon shock cord and a 36” diameter nylon cloth parachute. The kit no longer contains the fire blanket chute protector.

Hardware

The kit contains other hardware, such as rail buttons, an eye bolt, and a quick link to hold the nylon shock cord.

Miscellaneous Information

This kit is very flexible and can switch the payload bay into an electronics bay (e-bay) so that the rocket can use a dual-deployment recovery. Note that it is important to purchase a matching e-bay size so that it fits properly. Also, a recommended addition to purchase for this kit is a 54mm retainer.

Competitor: Ultimate (4.00" diameter)

The second competitor we selected was a cluster high-power rocket kit from Apogee Rockets called Ultimate [2]. This kit is \$144.47 and consists of the body tubes, nose cone, plywood fins, motor mount tubes, and the recovery system.

Body Tubes

The kit contains a heavy-duty, thick-walled paper body tube with pre-slotted fins. The total length of the rocket is 57.75" with a max diameter of 4.00" and a weight of 3.45lbs. The payload bay has an internal diameter and length of 3.8" and 10.75", respectively. It also comes with seven 29mm motor mount tubes for the cluster rocket.

Nose Cone

The nose cone is 11.8" long and made of heavy-duty polypropylene plastic. The nose cone also has a circular attachment for the nylon shock cord. The CP location is 50.17" from the tip of the nose cone.

Plywood Parts

The plywood components are made from premium $\frac{1}{8}$ " thick plywood. This kit contains six fins with a fin span of 14.00". It also has three centering rings that grip around the cluster of the 29mm motor mount tubes. A bulkhead is included as well.

Recovery System

The recovery system contains the nylon shock cord and a 42” rip-stop nylon parachute.

Miscellaneous Information

This rocket kit may be slightly different compared to the group’s rocket design because it contains multiple motors. This rocket can hold up to seven 29mm diameter motors per flight. The capable impulse the rocket can withstand is F to H class motors, and depending on the motor configuration, it can fly between 500 ft and 4,000 ft. Additionally, if the rocket is flown with lower class motors there is no need to add weight in the nose cone because the CP is already super far from the nose cone; however, if it is flown with a high-power sized motor then weight may need to be added in the nose cone. The payload bay for this kit is rather large, but the kit can be upgraded to a dual deployment by adding a 4.00” electronics bay.

Competitor: Vogel (2.00” diameter)

The third competitor we selected was a high-power rocket kit from Apogee Rockets called Vogel [3]. The cost of this rocket kit is \$85.59 and contains the body tubes, nose cone, fins, recovery system, and hardware.

Body Tubes

There are four main thick-walled body tubes due to it being a two-stage rocket kit. There is a coupler that is made from a material called “fishpaper,” which has a grey appearance and is very durable. This kit also has 29mm motor mount tubes. The payload bay's internal diameter and length are 1.95” and 9.3”, respectively. The max diameter is 2.00” and the total length of the rocket is 64.50”.

Nose Cone

The nose cone is made with high-impact polystyrene plastic and has a pre-made loop for the shock cord to connect to. The CP location is 52” from the tip of the nose cone.

Plywood Parts

The kit contains many plywood components: fins, bulkheads, centering rings, stuffer tube rings, and alignment rods. There are six fins in total (three fins for each stage), all having a fin span of 8.27".

Recovery System

The recovery system includes the aramid cords, shot cord, an 18" nylon parachute, and a 15" nylon parachute.

Miscellaneous Information

The total weight of the rocket is 1.06lbs. This kit can also fly without a payload bay by removing the payload bay and the coupler and directly connecting the nosecone to the body tube; however, this will change the stability of the rocket and may need to add weight to the nose of the rocket. This rocket can reach up to 2,000ft with D-F motors and can go even higher with G motors in the upper stage.

Competitor: HI-TECH (2.63" diameter)

The fourth competitor we selected was a high-power rocket kit from Apogee Rockets called Hi-Tech [4]. The cost of this rocket kit is \$101.76 and contains the body tubes, nose cone, fins, recovery system, and hardware.

Body Tubes

This kit has multiple body tubes that are made from thick-walled cardboard for extra strength. It also has pre-cut fin slots for the fins to insert into. The total length of the rocket is 49.75" with a max diameter of 2.63". The payload bay has an internal diameter of 2.56" and an internal length of 8". The kit also has a 38mm motor mount tube. The total weight of the rocket is 1.38lbs.

Nose Cone

The nose cone is 9" long and is made of plastic with an attachment area for the shock cord. The CP location is 41.71" from the tip of the nosecone.

Plywood Parts

The fins, bulkhead, and centering rings are made from 1/8" thick plywood. The fin span for the three fins is 9.50".

Recovery System

The recovery system that this kit has is a nylon shock cord and a 28" nylon parachute.

Hardware

The kit has an LL-25 launch lug and a shock cord mount.

Miscellaneous Information

Due to the rocket being very lightweight (~1.38lbs), it can fly over 6,000ft. The motor class suggestions are F to H. This rocket can also be used with a 29mm MMA-2 adapter, if needed.

Competitor: LOC Magnum- 3E (5.5" diameter)

The fifth competitor we selected was a high-power rocket kit from LOC Precision called LOC Magnum- 3E [5]. The cost of this kit is \$291.61 and contains the body tubes, fins, nose cone, and recovery system.

Body Tubes

The rocket kit comes with a 34" booster and a 34" payload section. The total length of the rocket is 91" and the diameter of the rocket is 5.5". The kit also includes a 75mm motor mount tube.

Plywood Parts

The fins for this kit are made with plywood and are $\frac{3}{8}$ " thick. The centering rings are made from plywood and are $\frac{1}{4}$ " thick.

Recovery System

The kit includes an SCM-4 $\frac{5}{8}$ " Y-harness 2x25' tubular nylon shock cord. The parachutes are ripstop nylon, with the main parachute being 78" and the drogue being 18".

Hardware

The kit contains RB-1500 rail buttons.

Miscellaneous Information

The total weight of the rocket is 9lbs. This kit also includes an electronics bay for dual deployment. There is a regular Magnum rocket kit as well, which is slightly smaller than the Magnum 3E. The Magnum does not contain the electronics bay for dual deployment. It is a perfect kit for an L1 or L2 certification flight, while the Magnum 3E is great for an L3 certification flight.

Competitor: Mad Dog (6" diameter)

The sixth competitor we selected was a high-power rocket kit from Madcow Rocketry called Mad Dog [6]. The cost of this rocket is \$738.00, and it contains the nose cone, body tubes, motor mount, fins, centering rings, and AV-bay lids.

Body Tubes

The body tube contains two body tubes: a 6" diameter filament-wound fiberglass air-frame that is 22" long and a 6" diameter filament-wound fiberglass slotted air-frame that is 44" long. Filament wound fiberglass is created by continuously winding fiberglass combined with a resin matrix around a rotating mandrel, this creates a customized product that is very

strong, lightweight, and durable. The kit also has a 6” diameter filament-wound fiberglass coupler that is 12” long. The motor mount tube is a 98mm fiberglass tube.

Nose Cone

The nose cone is a 6” fiberglass nose cone with an aluminum tip.

Other Fiberglass Parts

Almost all the other components in this kit are made of fiberglass. The kit has three beveled CNC-cut fiberglass 3/16” thick fins. There are also G10 fiberglass centering rings and CNC-machined stepped Av-bay lids.

Miscellaneous Information

Other information to note is that the total length of the rocket is slightly over 8ft tall. The built weight of the rocket is around 29lbs and also has a dual deployment system. The rocket can fly on motor classes from L to N class.

Our Rocket

After conducting research on other competitors in the market, several key design configurations must be considered for the group’s rocket design. Since the group has decided to include multiple payload deployments (Req. C.XI and F.XV.4), this requirement directly impacts the design considerations compared to the pre-made commercial kits.

When considering the airframe sizes and geometry, the competitor kits showed diameters ranging from 2.00” to 6.00”. Selecting a larger diameter (4.00” to 6.00”) provides advantages such as increased internal volume for payloads and electronics, easier installation of wiring and separation systems, and improved static stability at lower velocities (F.VII.1). The larger airframe also supports modularity (F.XI) and easier payload loading (F.XIV.1). However, drawbacks include increased drag, requiring higher total impulse and potentially more expensive

motor selections, as well as higher mass, which can reduce apogee (F.I.2), and larger recovery systems to maintain safe descent speeds (F.XII.1 and C.XXIV). Conversely, selecting a smaller diameter (2.00" to 3.00") reduces drag and can increase altitude performance for the same impulse, but sacrifices internal payload space and makes electronics integration more difficult due to the tighter tolerances (F.XIV.1). Overall, there are many takeaways to consider when selecting an airframe size, choosing a larger diameter may be more beneficial depending on the overall size of the payloads and depending on the number of payloads that will be deployed.

Material selection is another important consideration for major components such as the airframe, fins, bulkheads, payload/electronic bays, couplers, and centering rings. Cardboard offers advantages in cost and ease of fabrication (H.III, F.XX), making it a common choice for rockets. Another material that can be selected is fiberglass/carbon fiber. Fiberglass has high stiffness and strength, can handle higher dynamic pressure and thrust; however, it can be heavier, costlier, and requires more advanced fabrication. Since the group has a set budget requirement H.III, most likely the group will take the option that is less costly.

A key difference between most competitor kits and the group's design is payload configuration. While most of the kits are meant for single payload bay, the group's rocket must accommodate multiple payloads with sequential deployment (F.XV, F.XV.4). One possible design is modular stacked segmented bays, where each payload is separated by bulkheads or couplers. This configuration improves integration, testing, modularity (F.XI), and allows for different release mechanisms. Some drawbacks may include more materials like couplers and bulkheads which can increase the mass and possibly the wiring complexity. Another possible option is ejectable payload pods, where each payload has their own independent recovery system. While this design allows greater flexibility, it introduces more complexity by requiring

sequencing logic, separation assurance, and flight data tracking for each individual module (C.XVI.1 and F.XV.1).

Motor configuration was also a distinguishing factor among competitor kits, which often used motor classes ranging from F to N. This is important to note with each kit because each rocket can withstand a certain amount of impulse. Using one HPR motor simplifies thrust alignment and system integration (F.I.1), but limits flexibility once the mass and impulse are fixed. This is imperative to consider if there is an altitude requirement (F.I.2). Clustered motor configurations provide redundancy and higher thrust potential, but require reliable ignition sequencing and can induce a yaw moment if unevenly ignited (F.VII.1). There is also a two-stage design that can work as well since it enables higher altitude and can drop the booster in case mass needs to be reduced, but this requires complex staging coding and increases risk of failure.

In conclusion, commercial high-power kits offer valuable insight into baseline geometries, motor configurations, and recovery systems. However, the group's final rocket design must diverge to support multiple payload deployments, modular payload bays, and enhanced data acquisition systems (C.XVI, F.XIII, and C.XVII.1), all while not surpassing the set budget (H.III), structural integrity (F.XXI), and aerodynamic performance (F.II). These tradeoffs highlight that while the competitor kits are useful benchmarks, the group's design must be optimized for multi-payload operation for our specific goal.

Relationships Matrix

The translation of customer wants and needs into quantifiable technical specifications guarantees that each aspect of the rocket design aligns with user expectations about performance,

dependability, and safety. This connects qualitative objectives—such as “secure recovery,” “elevated altitude,” and “consistent flight”—with quantitative measurements that engineers may develop, evaluate, and validate. A relationship matrix is employed to illustrate and quantify these associations. Each customer's needs are cross-referenced with engineering parameters like altitude, descent rate, battery life, data sampling rate, and deployment schedule. This matrix employs weighted numerical values and correlation symbols to denote the magnitude of the effect of a certain engineering parameter on customer needs.

Relationship Strength	Value	Meaning
<u>Strong</u>	9	A strong direct impact between the engineering metric and user need
<u>Moderate</u>	3	A moderate but noticeable influence
<u>Weak</u>	1	A small or limited relationship
<u>None</u>	0	No meaningful relationship

Correlation Direction	Symbol	Meaning
<u>Positive</u>	+	Increasing this metric improves performance and meets a user need
<u>None</u>	•	No clear correlation
<u>Negative</u>	–	Increasing this metric harms performance or conflicts with a user need

The greater the number, the more important that statistic is to fulfilling mission objectives. For example, a "+9" under altitude indicates a strong positive relationship—improving altitude performance directly meets the requirement for maximum flight height. In contrast, a "-9" under descent speed indicates that increased speed significantly reduces recovery safety, indicating a trade-off that the team must carefully balance.

With this weighted approach, the design team may properly prioritize engineering effort. During design optimization, testing, and validation, relationships with a higher weight are given more attention. Negative connections reveal potential conflicts, such as when increasing cargo weight has a negative influence on ascent speed or altitude, and assist the team in making educated trade-offs early in the design process.

The relationship matrix also plays an important role in validation and verification. Validation assures that the design meets user expectations, such as safe and reliable recovery. Verification shows that quantitative requirements, such as a 10 Hz sampling rate or maximum of 11 m/s decent rate, are met through testing. By integrating these two functions, the matrix acts as a performance map, connecting each design variable to specific test results.

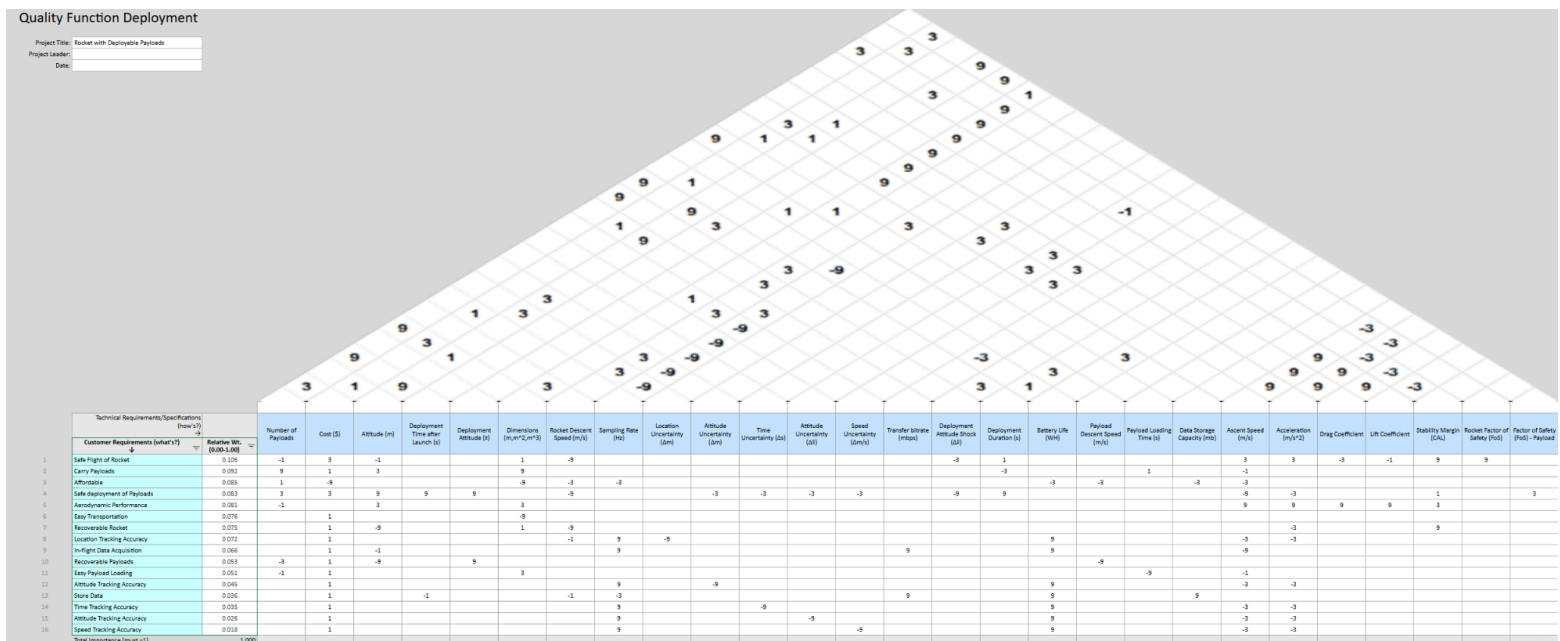


Figure 3. House of Quality (Relationships Matrix)

Technical Requirements/Specifications (how's?) ↑		↓																														
		Customer Requirements (what's?) ↓	Relative Wt. (0.00-1.00)	Number of Payloads		Altitude (m)	Cost (\$)	Deployment Time after Launch (s)	Deployment Attitude (x)	Dimensions (m,m^2,m^3)	Rocket Descent Speed (m/s)	Sampling Rate (Hz)	Location Uncertainty (Δm)	Altitude Uncertainty (Δm)	Time Uncertainty (Δs)	Attitude Uncertainty (Δx)	Speed Uncertainty (Δm/s)	Transfer bitrate (mbps)	Deployment Attitude Shock (Δx)	Deployment Duration (s)	Battery Life (Wh)	Payload Descent Speed (m/s)	Payload Loading Time (s)	Data Storage Capacity (mb)	Ascent Speed (m/s)	Acceleration (m/s^2)	Drag Coefficient	Lift Coefficient	Stability Margin (CAL)	Rocket Factor of Safety (FoS)	Factor of Safety (FoS) - Payload	
Safe Flight of Rocket		0.106	-1	3	-1			1	-9											-3	1					3	3	-3	-1	9	9	
Carry Payloads		0.092	9	1	3			9												-3			1			-1						
Affordable		0.085	1	-9				-9	-3	-3											-3	-3			-3	-3						
Safe deployment of Payloads		0.083	3	3	9	9	9		-9			-3	-3	-3	-3				-9	9					-9	-3			1		3	
Aerodynamic Performance		0.081	-1		3			3																	9	9	9	9	3			
Easy Transportation		0.076		1				-9																								
Recoverable Rocket		0.075		1	-9			1	-9																	-3			9			
Location Tracking Accuracy		0.072		1					-1	9	-9										9				-3	-3						
In-flight Data Acquisition		0.066		1	-1					9								9			9				-9							
Recoverable Payloads		0.053	-3	1	-9		9															-9										
Easy Payload Loading		0.051	-1	1				3															-9		-1							
Altitude Tracking Accuracy		0.045		1						9		-9									9				-3	-3						
Store Data		0.036		1		-1			-1	-3								9			9			9								
Time Tracking Accuracy		0.035		1						9				-9							9				-3	-3						
Attitude Tracking Accuracy		0.026		1						9					-9						9				-3	-3						
Speed Tracking Accuracy		0.018		1						9							-9				9				-3	-3						

Figure 4. Relationships Matrix (Without Pyramid)

Critical Performance Targets

Through the analysis of the House of Quality and the Engineering Requirements matrix, key performance characteristics have been identified as drivers of system success. Each critical parameter has been assigned preliminary target values, tolerances, and validation methods to ensure performance verification for each requirement. These parameters determine whether the system meets its operational objectives, cost constraints, and reliability expectations. The other parameters are not as crucial to the system. The main verification methods are inspection, analysis, and testing. Inspection is an examination that utilizes the basic senses and simple measurements. Analysis uses calculations and computer models for verification. Testing involves placing the system under experimental conditions to verify benchmark conditions. Many of the specifications will be verified by inspection and testing.

A critical factor of the system design is the number of payloads. The scope of the project plans to have two payloads deployed during descent in sequence with a third payload deployed after landing. The rocket's and payloads' maximum descent speed is 11 meters per second, as defined by TRA regulations. The rocket should launch in suitable weather conditions. Unpredictable weather, such as high-speed winds, can alter the planned trajectory of the rocket and its payloads. Clouds can cause precipitation, wind, lightning, and visibility issues. Cloud coverage should not exceed 50%. The rocket should not be launched if there are cumulus-type clouds with tops higher than 41 degrees Fahrenheit due to the potential for precipitation.

For the ground deployment, the rocket should be oriented horizontally on the ground, with the roll, pitch, and yaw being 0 degrees. Due to the shape of the rocket being cylindrical, the tolerance for the roll is 180 degrees. The pitch and yaw have a tolerance of 10 degrees. The plan for the ground deployment takes into account that having the rocket be vertical is not feasible.

The minimum altitude that needs to be achieved is 1600 meters, and this can be tested through simulations and the motor impulse. The actual altitude will need to be measured through a test flight. There must be ample time for the deployment, so a high altitude was chosen.

The deployment should have a maximum time duration of 10 seconds. The rocket's orientation when deploying payloads should ideally be perpendicular to the ground or 90 degrees. Wind can cause the rocket not to have an ideal attitude; hence, the tolerance is 20 degrees. The deployment method should not create an attitude shock of more than 15 degrees. A higher attitude shock can cause the rocket to veer from its initial landing zone.

The overall cost of the rocket will be under \$700, but that is still open to debate due to component costs and the lack of an overall budget. The budget will be referenced as each component and material is selected.

Engineering Requirements

After team deliberation of customer wants and needs, competing designs, and identification of critical performance parameters, the engineering requirements were drafted. Each requirement is sourced from one of the previously listed idea banks. For each requirement, an intended target value or tolerance was chosen such that the team could validate whether the requirement was met using either an inspection, test, or analysis. In the case of most requirements listed below, a test will refer to a launch of the rocket and evaluating its performance, an analysis will refer to the use of software to model the launch or process of a specific subsystem, and an inspection will refer to the given manufacturer specification for the component.

Table 1. Engineering Requirements and Preliminary Targets

Reference Number	Requirement	Target Value / Tolerances	Validation Method
C.I.1	Motor should be powerful enough to conduct the launch	320-640Ns	Inspection
C.VII	Rocket must be compatible with the launch system	Has launch rail guides	Inspection
C.VIII	Battery must power the electronics for the entire duration of the launch	7.5Wh min	Test
C.XI	Must carry a set amount of payloads	3	Inspection
C.XVI	Storage must be large enough to log all flight data	10GB min	Inspection
C.XVI.1	Location uncertainty must be within a certain tolerance	10m max	Analysis
C.XVI.1	Altitude uncertainty must be within a certain tolerance	10m max	Analysis
C.XVI.1	Time uncertainty must be within a certain tolerance	1s max	Analysis
C.XVI.1	Attitude uncertainty must be within a certain tolerance	15 deg max	Analysis
C.XVI.1	Speed uncertainty must be within a certain tolerance	20m/s max	Analysis
C.XVII.1	Rocket should transfer data quick enough to get real-time measurements	TBD	Test
C.XXIV	Payload should slow on descent to a safe landing speed	11m/s max	Test
F.I.1	Rocket must reach a minimum ascent speed	20m/s min	Test
F.I.2	Must reach a minimum altitude	1600m min	Test
F.II	Drag Coefficient of the rocket body on launch should be under a set maximum	0.8 max	Analysis
F.III	Rocket must be launched in suitable weather conditions	Cumulus Cloud Temperature: 41F max Cloud Coverage: 5/10 max Precipitation: None	Inspection
F.IV	Motor shall not be ignited by use of a mercury switch or roller switch		Inspection
F.VI	Rocket should be easy to assemble	30min max	Test
F.VII.1	Stability Margin must be within a set range	1.5-2 $\pm$ 0.2	Analysis
F.XI	Rocket must be modular	2 sections minimum	Inspection
F.XII	Rocket should be horizontal after landing	(0,0,0) $\pm$ (180,10,10)	Test
F.XII.1	Rocket must slow on descent to a safe landing speed	11m/s max	Test
F.XIII	Electronics must have a sampling rate quick enough to get accurate data	100Hz $\pm$ 10Hz	Test
F.XIV.1	Payloads should be loaded within a set time	1m max for each payload	Test
F.XV	Deployments must be within a maximum time duration	10s max	Test
F.XV.1	Rocket should be oriented properly when deploying payloads	(0,90,0) $\pm$ (180,20,20)	Test
F.XV.1	Deployment should create an attitude shock within a reasonable limit	15 deg max	Test
F.XV.4	Payload deployments should be sequential with a minimum time spacing	3s min	Test
F.XVI.1	Rocket must be large enough to house all components	Dependent on Motor/Payload Sizing	Inspection
F.XX	Rocket must be produced with available machines	Tolerancing: TBD	Inspection
F.XXI	Rocket's factor of safety must be within a set range	2-4 $\pm$ 0.5	Analysis

F.XXI	The payload deployment's factor of safety must be within a set range	$2-4 \pm 0.5$	Analysis
F.XXII	Rocket must withstand dynamic pressure	$6\text{kPa} \pm 1\text{kPa}$	Test
H.III	Costs should stay below a set target	\$700 max	Inspection

As shown by the ‘Reference Number’ column, all of the proposed engineering requirements were matched to either the component decomposition (C.#) or functional decomposition (F.#), with the one exception of cost, which did not match either decomposition. This value came solely from customer wants and needs as explored earlier in the House of Quality (H.#).

C.I — The selected motor must be powerful enough to conduct the launch with an impulse range of 320 Ns - 640 Ns. The total impulse of the selected motor will be determined by the manufacturer's data. The target impulse range was set to conform to the Level 1 High Powered Rocket Certification standards.

C.VII — Since the launch site provides launch rails, the rocket shall be able to properly launch with its equipment. There must be launch rail guides to ensure the rocket can be placed on the rails and can launch without interference.

C.VIII — The electronics onboard that control timings of payload deployments, parachute ejection, and data collection require a battery to run. That battery must be powerful enough to support the entire launch-to-landing cycle. The target value of 7.5Wh was chosen as a medium-lifespan battery that will be more than capable of fulfilling the above

needs. This will be validated by testing the battery lifespan over an estimated launch period.

C.XI — The rocket will deploy 3 separate payloads during the launch. This can be validated with a simple inspection of the number of payloads present in the rocket.

C.XVI — Storage space must be large enough to retain all collected data acquired during flight, the target value being 10 GB. The data being collected includes location, altitude, time, attitude, and speed; the uncertainty of each data point must fall within certain ranges. Location and altitude uncertainties are limited to 10 meters, time uncertainty is limited to 1 second, attitude uncertainty is limited to 15 degrees, and speed uncertainty is limited to 20 meters per second.

C.XVII — During the launch, the team intends to obtain real-time telemetry from the rocket. The method of communication must be quick enough and have enough bandwidth to support data from all sensors on the rocket. The required speed and bandwidth have yet to be decided and will depend on the amount of data collected from the sensors. Once chosen, the target can be validated by testing the data loss of the remote signal during a simulated launch.

C.XXIV — To conform with regulations at the proposed launch site, all payloads ejected from the rocket in flight must land with a maximum descent speed of 11 m/s. Tests will be performed to evaluate the landing speed of the deployed payloads.

F.I — The rocket must meet a minimum ascent speed and reach a minimum altitude to ensure a safe launch and deployment sequence. The minimum target ascent speed is 20 meters per second, and the minimum target altitude is 1,600 meters.

F.II — The total drag coefficient of the rocket body should be under 0.8 to ensure the minimum altitude and ascent speed are reached efficiently.

F.III — The rocket will be built to withstand certain weather conditions. The intended launch site of Palm Bay sets forth basic standards of when launches can take place. Cloud cover must be no more than five-tenths, it must not be raining, the rocket cannot be flown directly into a cloud, and cloud temperature must be no more than 41°F. These conditions shall be validated by the launch site staff before each attempt.

F.IV — Per TRA guidelines, the motor must not be ignited by use of a mercury switch or a roller switch to prevent incidental activations.

F.VI — The rocket should be easy to assemble with a maximum target assembly time of 30 minutes. This is to allow the team plenty of time for last-minute checks and to avoid unnecessary assembly delays before flight.

F.VII — The stability margin of the rocket must be between 1.5 and 2 cal. This ensures the vehicle remains on its proper and predictable trajectory to reach the required altitude.

Stability margin will be calculated using a computer-modelling program.

F.XI — To ensure that the rocket can be safely transported and stored when not in flight, the team requires it to be modular. These sections refer only to partitions of the body, not the nose cone or fins. The required minimum is 2 sections; however, the usage of stage separation to deploy payloads may increase the number of sections used.

F.XII — Since the third payload's deployment after flight can be affected by the attitude of the rocket once it's on the ground, the preferred orientation of the rocket after landing is horizontal with respect to the ground.

F.XIII — Electronics that are responsible for acquiring in-flight data must have a target sampling rate of $100\text{Hz} \pm 10\text{Hz}$. This is to guarantee data accuracy.

F.XIV — Payloads should be loaded fairly quickly – within 1 minute for each payload – for the ease of the team and to avoid unnecessary delays before flight.

F.XV — Since the team wants to deploy two payloads while airborne, both after the rocket reaches apogee, each deployment cycle must be done relatively quickly. To ensure both can be done before the rocket is too low, the team wants each deployment cycle to take no longer than 10 seconds, and each deployment must have at least 3 seconds between

them so the rocket has time to regain stability. While deploying these payloads, the rocket must be oriented vertically under a drogue, with a margin for error of a 20° lean. While the payload is being deployed, the deployment shock must tilt the rocket no more than 15° so that it does not become unstable, and so that the second deployment and later landing will not be affected. All of these conditions will be validated with a test launch or simulated deployment test.

F.XVI — The rocket needs to house all of its components, including the payloads, electronics, and deployment mechanisms. No set size requirement is set yet, as many of the final sizes of these components will not be known until after the ideation process.

F.XX — While manufacturing the rocket, the team is limited to available machines at UCF and their tolerances. The team must design their component tolerances to be no more precise than the machines can ensure. These values will come later once the machines have been identified.

F.XXI — Both the rocket and payload deployment mechanisms will be manufactured with a structural factor of safety of 2-4.

F.XXII — During launch, rockets experience dynamic pressure above the standard atmospheric pressure. As a baseline, this rocket must be able to withstand a dynamic pressure of up to 6kPa, which will be calculated via onboard sensors during a launch.

H.III — The team is working on a budget for this project, as requested in our customer requirements. After speaking with our customer (Gary & Kurt), the team is assuming a budget cap of \$700 as a starting point for the ideation and manufacturing process.

The analysis of customer wants and needs, engineering requirements, and competitor benchmarks has positioned the team to start the ideation phase with clear, measurable targets. Guided by the performance criteria defined above, the rocket can be designed with focus on safety, reliability, and modular payload integration within the provisional budget set forth. These requirements will remain an integral part of the design process, and will be how the team decides whether the project was actually a success at the completion of the building and testing phases.

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